

```
import numpy as np
import matplotlib.pyplot as plt

# =====
# 1. Bar Plot
# =====

x = np.array(["A", "B", "C", "D"])
y = np.array([10, 20, 15, 25])

plt.bar(x, y)
plt.title("Bar Plot")
plt.xlabel("Categories")
plt.ylabel("Values")
plt.show()

# =====
# 2. Line Plot
# =====

x = np.array([1, 2, 3, 4, 5])
y = np.array([10, 20, 15, 25, 30])

plt.plot(x, y)
plt.title("Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()

# =====
# 3. Pie Plot
# =====

x = np.array([30, 20, 25, 25])
labels = ["Python", "Java", "C++", "HTML"]

plt.pie(x, labels=labels, autopct="%1.1f%%")
plt.title("Pie Plot")
plt.show()

# =====
# 4. Scatter Plot
# =====

x = np.array([1, 2, 3, 4, 5])
y = np.array([5, 10, 15, 20, 25])

plt.scatter(x, y)
plt.title("Scatter Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()

# =====
# 5. Histogram
# =====

x = np.array([10, 20, 20, 30, 30, 30, 40, 40, 50])

plt.hist(x, bins=5)
plt.title("Histogram")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()

# =====
# 6. Box Plot
# =====

x = np.array([10, 20, 25, 30, 35, 40, 50])

plt.boxplot(x)
plt.title("Box Plot")
plt.ylabel("Values")
plt.show()
```




